Precise Zero-Shot Dense Retrieval without Relevance Labels

Luyu Gao[*†] **Xueguang Ma**[*‡] **Jimmy Lin**[‡] **Jamie Callan**[†]
[†]Language Technologies Institute, Carnegie Mellon University
[‡]David R. Cheriton School of Computer Science, University of Waterloo
{luyug, callan}@cs.cmu.edu, {x93ma, jimmylin}@uwaterloo.ca

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Abstract

While dense retrieval has been shown effective and efficient across tasks and languages, it remains difficult to create effective fully zero-shot dense retrieval systems when no relevance label is available. In this paper, we recognize the difficulty of zero-shot learning and encoding relevance. Instead, we propose to pivot through Hypothetical Document Embeddings (HyDE). Given a query, HyDE first zero-shot instructs an instruction-following language model (e.g. InstructGPT) to generate a *hypothetical* document. The document captures relevance patterns but is unreal and may contain false details. Then, an unsupervised contrastively learned encoder (e.g. Contriever) encodes the document into an embedding vector. This vector identifies a neighborhood in the corpus embedding space, where similar *real* documents are retrieved based on vector similarity. This second step ground the generated document to the actual corpus, with the encoder's dense bottleneck filtering out the incorrect details. Our experiments show that HyDE significantly outperforms the state-of-the-art unsupervised dense retriever Contriever and shows strong performance comparable to fine-tuned retrievers, across various tasks (e.g. web search, QA, fact verification) and languages (e.g. sw, ko, ja).[1]

1 Introduction

Dense retrieval (Lee et al., 2019; Karpukhin et al., 2020), the method of retrieving documents using semantic embedding similarities, has been shown successful across tasks like web search, question answering, and fact verification. A variety of methods such as negative mining (Xiong et al., 2021; Qu et al., 2021), distillation (Qu et al., 2021; Lin et al., 2021b; Hofstätter et al., 2021) and task-specific pre-training (Izacard et al., 2021; Gao and Callan, 2021; Lu et al., 2021; Gao and Callan, 2022; Liu and Shao, 2022) have been proposed to improve the effectiveness of supervised dense retrieval models.

On the other hand, zero-shot dense retrieval still remains difficult. Many recent works consider the alternative transfer learning setup, where the dense retrievers are trained on a high-resource dataset and then evaluated on queries from new tasks. The MS-MARCO collection (Bajaj et al., 2016), a massive judged dataset with a large number of judged query-document pairs, is arguably the most commonly used. As argued by Izacard et al. (2021), in practice, however, the existence of such a large dataset cannot always be assumed. Even MS-MARCO restricts commercial use and cannot be adopted in a variety of real-world search scenarios.

In this paper, we aim to build effective fully zero-shot dense retrieval systems that require **no relevance** supervision, work out-of-box and generalize across tasks. As supervision is not available, we start by examining self-supervised representation learning methods. Modern deep learning enables two distinct learning algorithms. At the token level, generative large language models (LLM) pre-trained on large corpus have demonstrated strong natural language understanding (NLU) and generation (NLG) capabilities (Brown et al., 2020; Chen et al., 2021; Rae et al., 2021; Hoffmann et al., 2022; Thoppilan et al., 2022; Chowdhery et al., 2022). At the document level, text (chunk) encoders pre-trained with contrastive objectives learn to encode document-document similarity into inner-product (Izacard et al., 2021; Gao and Callan, 2022). On top of these, one extra insight into LLM is borrowed: the LLMs further trained to follow instructions can *zero-shot* generalize to diverse unseen instructions (Ouyang et al., 2022; Sanh et al., 2022; Min et al., 2022; Wei et al., 2022). Ouyang et al. (2022) show that with a small amount of data, GPT-3 (Brown et al., 2020) models can be aligned

[*] Equal contribution.
[1]No models were trained or fine-tuned in making this preprint. Our open source code is available at https://github.com/texttron/hyde.

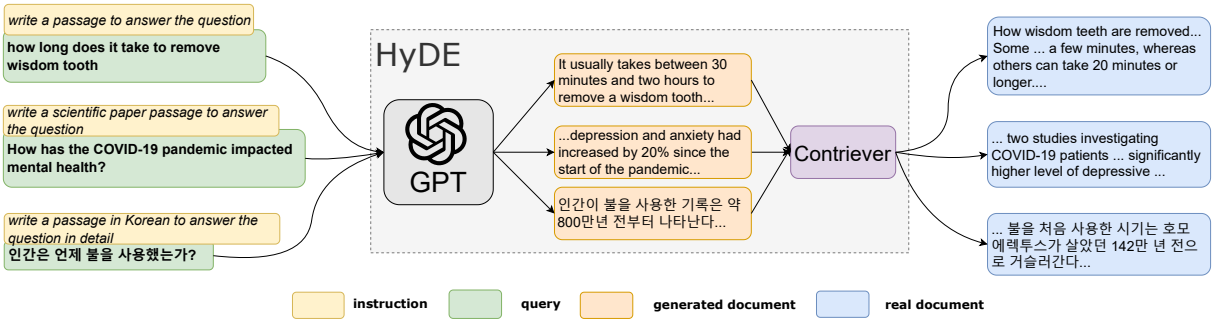

Figure 1: An illustration of the HyDE model. Documents snippets are shown. HyDE serves all types of queries without changing the underlying GPT-3 and Contriever/mContriever models.

to human intent to follow instructions.

With these ingredients, we propose to pivot through Hypothetical Document Embeddings (HyDE), and decompose dense retrieval into two tasks, a generative task performed by an instruction-following language model and a document-document similarity task performed by a contrastive encoder (Figure 1). First, we feed the query to the generative model and instruct it to "write a document that answers the question", i.e. a hypothetical document. We expect the generative process to capture "relevance" by giving an example; the generated document **is not** real, can contain factual errors but is like a relevant document. In the second step, we use an unsupervised contrastive encoder to encode this document into an embedding vector. Here, we expect the encoder's dense bottleneck to serve a lossy compressor, where the extra (hallucinated) details are filtered out from the embedding. We use this vector to search against the corpus embeddings. The most similar *real* documents are retrieved and returned. The retrieval leverages document-document similarity encoded in the inner-product during contrastive training. Note that, interestingly, with HyDE factorization, the query-document similarity score is no longer explicitly modeled nor computed. Instead, the retrieval task is cast into two NLU and NLG tasks.

HyDE appears unsupervised. **No** model is trained in HyDE: both the generative model and the contrastive encoder remain intact. Supervision signals were only involved in instruction learning of our backbone LLM.

In our experiments, we show HyDE using InstructGPT (Ouyang et al., 2022) and Contriever (Izacard et al., 2021) as backbone models significantly outperforms the previous state-of-the-art Contriever-only zero-shot no-relevance system on 11 queries sets, covering tasks like Web Search, Question Answering, Fact Verification and languages like Swahili, Korean, Japanese.

2 Related Works

Dense Retrieval (Lee et al., 2019; Karpukhin et al., 2020) has been extensively studied after the emergence of pre-trained Transformer language models (Devlin et al., 2019). Researchers studied the metric learning problems, such as training loss (Karpukhin et al., 2020) and negative sampling (Xiong et al., 2021; Qu et al., 2021), and also introduced distillation (Qu et al., 2021; Lin et al., 2021b; Hofstätter et al., 2021). Later works studied the second stage pre-training of language model specifically for retrieval (Izacard et al., 2021; Gao and Callan, 2021; Lu et al., 2021; Gao and Callan, 2022; Liu and Shao, 2022).

The popularity of dense retrieval can be partially attributed to the rich and successful research in very efficient minimum inner product search (MIPS) at very large (billion) scales (Johnson et al., 2017).

Instructions-Following Language Models Soon after the emergence of LLMs, several groups of researchers discover that LLMs trained on data consisting of instructions and their execution can zero-shot generalize to perform new tasks with new instructions (Ouyang et al., 2022; Sanh et al., 2022; Min et al., 2022; Wei et al., 2022). This can be done by standard supervised sequence-to-sequence learning or more effectively with reinforcement learning (Ouyang et al., 2022).

Concurrent to us, Asai et al. (2022) studied "Task-aware Retrieval with Instructions". They *fine-tuned dense encoders* that can also encode task-specific instruction prepended to query. In comparison, we use an unsupervised encoder and handle different tasks and their instruction with an

instruction following generative LLM, as described above.

Zero-Shot Dense Retrieval The tasks of zero-shot (dense) retrieval are arguably empirically defined by Thakur et al. (2021) for the neural retrieval community. Their BEIR benchmark consists of diverse retrieval tasks. The paper and many follow-up research generally consider the `Transfer Learning` setup where the dense retriever is first learned using a diverse and richly supervised corpus and query collection, namely MS-MARCO (Thakur et al., 2021; Wang et al., 2022; Yu et al., 2022).

However, as stated by Izacard et al. (2021), such a large collection can rarely be assumed. In this paper, therefore, we study the problem of building effective dense retrieval systems without relevance labels. Similar to Izacard et al. (2021), we also do not assume access to the test time corpora for training. This is a more realistic setup and prevents over-engineering on the test corpora.

By the definition in Sachan et al. (2022), our setup can be roughly considered as **"unsupervised"**. Strictly, as with Sachan et al. (2022), the only supervision resides in the LLM, in the processing of learning to follow instructions.

Generative Retrieval Generative search is a new class of retrieval methods that use neural generative models as search indices (Metzler et al., 2021; Tay et al., 2022; Bevilacqua et al., 2022; Lee et al., 2022). These models use (constrained) decoding to generate document identifiers, such as id and sub-string, which map directly to *real* documents. They have to go through special training procedures over relevance data; effective search may also need to use novel forms of search indices (Bevilacqua et al., 2022; Lee et al., 2022). In comparison, our method uses the standard MIPS index and requires no training or training data. Our generative model produces an intermediate hypothetical document to be fed into a dense encoder, instead of a real document.

3 Methodology

In this section, we first formally define the problem of (zero-shot) dense retrieval. Then we will introduce how `HyDE` is designed to solve it.

3.1 Preliminaries

Dense retrieval models similarity between query and document with inner product similarity. Given a query q and document d, it uses two encoder function enc_q and enc_d to map them into d dimension vectors $\mathbf{v_q}, \mathbf{v_d}$, whose inner product is used as similarity measurement.

$$\text{sim}(\text{q}, \text{d}) = \langle \text{enc}_q(\text{q}), \text{enc}_d(\text{d}) \rangle = \langle \mathbf{v_q}, \mathbf{v_d} \rangle \quad (1)$$

For zero-shot retrieval, we consider L query sets $Q_1, Q_2, ..., Q_L$ and their corresponding search corpus, document sets $D_1, D_2, ..., D_L$. Denote the j-th query from i-th set query set Q_i as q_{ij}. We need to fully define mapping *functions* enc_q and enc_d without access to any query set Q_i, document set D_i, or any relevance judgment r_{ij}.

The difficulty of zero-shot dense retrieval lies precisely in Equation 1: it requires learning of two embedding functions (for query and document respectively) into the *same* embedding space where inner product captures *relevance*. Without relevance judgments/scores to fit, learning becomes intractable.

3.2 HyDE

`HyDE` circumvents the aforementioned learning problem by performing search in document-only embedding space that captures document-document similarity. This can be easily learned using unsupervised contrastive learning (Izacard et al., 2021; Gao et al., 2021; Gao and Callan, 2022). We set document encoder enc_d directly as a contrastive encoder enc_{con}.

$$f = \text{enc}_d = \text{enc}_{\text{con}} \quad (2)$$

This function is also denoted as f for simplicity. This unsupervised contrastive encoder will be shared by all incoming document corpus.

$$\mathbf{v_d} = f(d) \quad \forall d \in D_1 \cup D_2 \cup ... \cup D_L \quad (3)$$

To build the query vector, we consider in addition an instruction following LM, InstructLM. It takes a query q and a textual instruction INST and follows them to perform the task specified by INST. For simplicity, denote,

$$g(q, \text{ INST}) = \text{InstructLM}(q, \text{ INST}) \quad (4)$$

Now we can use g to map queries to "hypothetical" documents by sampling from g, setting INST

to be "`write a paragraph that answers the question`". The generated document *is not* real, can and is likely to be ungrounded factually (Brown et al., 2020; Thoppilan et al., 2022). We *only* require it to capture relevance pattern. This is done by generating documents, i.e. providing examples. Critically, here we **offload** relevance modeling from representation learning model to an NLG model that generalizes significantly more easily, naturally, and effectively (Brown et al., 2020; Ouyang et al., 2022). Generating examples also replaces explicit modeling of relevance scores.
We can now encode the generated document using the document encoder f. Write,

$$\mathbb{E}[\mathbf{v}_{q_{ij}}] = \mathbb{E}[f(g(q_{ij}, \text{INST}_i))] \tag{5}$$

Formally, g defines a probability distribution based on the chain rule. In this paper, we simply consider the expectation value, assuming the distribution of $\mathbf{v}_{q_{ij}}$ is uni-modal, i.e. the query is not ambiguous. The study of ambiguous queries and diversity is left to future work. We estimate Equation 5 by sampling N documents from g, $[\hat{d_1}, \hat{d_2}, ..., \hat{d_N}]$.

$$\hat{\mathbf{v}}_{q_{ij}} = \frac{1}{N} \sum_{\hat{d_k} \sim g(q_{ij}, \text{INST}_i)} f(d_k) \tag{6}$$

$$= \frac{1}{N} \sum_{k=1}^{N} f(\hat{d_k}) \tag{7}$$

We also consider the query as a possible hypothesis,

$$\hat{\mathbf{v}}_{q_{ij}} = \frac{1}{N+1}[\sum_{k=1}^{N} f(\hat{d_k}) + f(q_{ij})] \tag{8}$$

Inner product is computed between $\hat{\mathbf{v}}_{q_{ij}}$ and the set of all document vectors $\{f(d)|d \in D_i\}$. The most similar documents are retrieved. Here the encoder function f serves as a lossy compressor that outputs dense vectors, where the extra details are filtered and left out from the vector. It further grounds the hypothetical vector to the actual corpus and the real documents. The full `HyDE` system is illustrated in Figure 1.

4 Experiments

4.1 Setup

Implementation We implement `HyDE` using `InstructGPT`, a GPT-3 model from the instruct series (`text-davinci-003`; Ouyang et al. (2022)) and `Contriever` models (Izacard et al., 2021). We sample from `InstructGPT` using the OpenAI playground default temperature of 0.7 for open-ended generations. We use the English-only `Contriever` model for English retrieval tasks and multilingual `mContriever` for non-English tasks. We conducted retrieval experiments with the Pyserini toolkit (Lin et al., 2021a).

Datasets We consider web search query sets TREC DL19 (Craswell et al., 2020a) and DL20 (Craswell et al., 2020b); they are based on the MS-MARCO dataset (Bajaj et al., 2016). We also use a diverse collection of 6 low-resource datasets from the BEIR dataset (Thakur et al., 2021). For non-English retrieval, we consider Swahili, Korean, Japanese, and Bengali from the Mr.Tydi dataset (Zhang et al., 2021).

We use different instructions for each dataset. They share a similar structure but have different quantifiers to control the exact form of the generated hypothetical documents. These instructions can be found in subsection A.1.

Compared Systems Contriever models, `Contriever` and `mContriever`, serve as our major baseline. They are trained using unsupervised contrastive learning. `HyDE` retrievers share the *exact* same embedding spaces with them. The only difference is how the query vector is built. These comparisons allow us to easily examine the effect of `HyDE`. The classical heuristic-based lexical retriever BM25 is also included.

Several systems that involve fine-tuning on massive *relevance* data are also included as references. We consider models fine-tuned on MS-MARCO and transferred, DPR and ANCE, from the BEIR paper. For multilingual, we include the mDPR model from Mr.Tydi paper and MS-MARCO fine-tuned mBERT and XLM-R from the Contriever paper. We also include the state-of-the-art transfer learning models: `Contriever` and `mContriever` fine-tuned on MS-MARCO, denoted `Contriever`$^{\text{FT}}$ and `mContriever`$^{\text{FT}}$. These models have run through the state-of-the-art retrieval model training pipeline that involves second-stage retrieval-specific pre-training (Lee et al., 2019) and a few rounds of fine-tuning (Qu et al., 2021); they should be considered empirical upper bounds.

4.2 Web Search

In Table 1, we show retrieval results on TREC DL19 and TREC DL20. We see `HyDE` bring sizable improvements to `Contriever` across the board for

	DL19			DL20		
	map	ndcg@10	recall@1k	map	ndcg@10	recall@1k
w/o relevance judgement						
BM25	30.1	50.6	75.0	28.6	48.0	78.6
Contriever	24.0	44.5	74.6	24.0	42.1	75.4
HyDE	**41.8**	**61.3**	**88.0**	**38.2**	**57.9**	**84.4**
w/ relevance judgement						
DPR	36.5	62.2	76.9	41.8	**65.3**	81.4
ANCE	37.1	**64.5**	75.5	40.8	64.6	77.6
Contriever^{FT}	41.7	62.1	83.6	**43.6**	63.2	**85.8**

Table 1: Results for web search on DL19/20. Best performing w/o relevance and overall system(s) are marked **bold**. DPR, ANCE and Contriever^{FT} are in-domain *supervised* models that are finetuned on MS MARCO training data.

	Scifact	Arguana	Trec-Covid	FiQA	DBPedia	TREC-NEWS
			nDCG@10			
w/o relevance judgement						
BM25	67.9	39.7	**59.5**	23.6	31.8	39.5
Contriever	64.9	37.9	27.3	24.5	29.2	34.8
HyDE	**69.1**	**46.6**	59.3	**27.3**	**36.8**	**44.0**
w/ relevance judgement						
DPR	31.8	17.5	33.2	29.5	26.3	16.1
ANCE	50.7	41.5	**65.4**	30.0	28.1	38.2
Contriever^{FT}	67.7	44.6	59.6	**32.9**	**41.3**	42.8
			Recall@100			
w/o relevance judgement						
BM25	92.5	93.2	**49.8**	54.0	46.8	44.7
Contriever	92.6	90.1	17.2	56.2	45.3	42.3
HyDE	**96.4**	**97.9**	41.4	**62.1**	**47.2**	**50.9**
w/ relevance judgement						
DPR	72.7	75.1	21.2	34.2	34.9	21.5
ANCE	81.6	93.7	45.7	58.1	31.9	39.8
Contriever^{FT}	94.7	97.7	40.7	**65.6**	**54.1**	49.2

Table 2: Low resource tasks from BEIR. Best performing w/o relevance and overall system(s) are marked **bold**.

both precision-oriented and recall metrics. While unsupervised Contriever can underperform the classical BM25 approach, HyDE outperforms BM25 by large margins.

HyDE remains competitive even when compared to fine-tuned models. Note that TREC DL19/20 are search tasks defined on MS-MARCO and there, all the fine-tuned models are richly *supervised*. On TREC DL19, HyDE shows comparable map and ndcg@10 to Contriever^{FT} and best recall@1k. On DL20, HyDE gets around 10% lower map and ndcg@10 than Contriever^{FT} and similar recall@1k. The ANCE model shows better ndcg@10 numbers than HyDE but lower recall, suggesting it may be biased to a subset of queries and/or relevant documents.

4.3 Low Resource Retrieval

In Table 2, we show retrieval results on low-resource tasks from BEIR. Similar to web search, HyDE again brings sizable improvements to Contriever across the board in terms of both ndcg and recall. HyDE is only outperformed by BM25 on one dataset, TREC-Covid but with a tiny 0.2 margin; in comparison, the underlying Contriever underperforms by more than 50%.

We also observe HyDE demonstrates strong performance compared to fine-tuned models. HyDE generally shows better performance than ANCE and DPR, even though the two are fine-tuned on MS-MARCO and ANCE also involves some sophisticated hard negative techniques. Contriever^{FT} shows performance advantages on FiQA and DBPedia. These involve retrieval of financial posts or entities respectively. We believe the performance difference can be attributed to the

	Swahili	Korean	Japanese	Bengali
w/o relevance judgement				
BM25	38.9	28.5	21.2	**41.8**
mContriever	38.3	22.3	19.5	35.3
HyDE	**41.7**	**30.6**	**30.7**	41.3
w/ relevance judgement				
mDPR	7.3	21.9	18.1	25.8
mBERT	37.4	28.1	27.1	35.1
XLM-R	35.1	32.2	24.8	41.7
mContriever^{FT}	**51.2**	**34.2**	**32.4**	**42.3**

Table 3: MRR@100 on Mr.Tydi. Best performing w/o relevance and overall system(s) are marked **bold**.

Model	DL19	DL20
Contriever	44.5	42.1
Contriever^{FT}	62.1	63.2
HyDE		
w/ Contriever		
w/ Flan-T5 (11b)	48.9	52.9
w/ Cohere (52b)	53.8	53.8
w/ GPT (175b)	**61.3**	**57.9**
w/ Contriever^{FT}		
w/ Flan-T5 (11b)	60.2	62.1
w/ Cohere (52b)	61.4	63.1
w/ GPT (175b)	**67.4**	**63.5**

Table 4: NDCG@10 on TREC DL19/20. Effect of changing different instruction LMs and using fine-tuned encoder. Best w/o relevance and overall models are marked **bold**.

under-specification of the instruction; more elaborative instructions may help.

4.4 Multilingual Retrieval

Multilingual setup poses several additional challenges to HyDE. The small-sized contrastive encoder gets saturated as the number of languages scales (Conneau et al., 2020; Izacard et al., 2021). Meanwhile, our generative LLM faces an opposite issue: with languages of not as high resource as English or French, the high capacity LLM can get under-trained (Hoffmann et al., 2022).

Nevertheless, in Table 3, we still find HyDE able to improve the mContriever model. It can outperform non-Contriever models fine-tuned on and transferred from MS-MARCO. On the other hand, we do observe some margins between HyDE and fine-tuned mContriever^{FT}. Since HyDE and mContriever^{FT} use similar contrastive encoders, we hypothesize this is because the non-English languages we considered are under-trained in both pre-training and instruction learning stages.

5 Analysis

The generative LLM and contrastive encoder make up the backbone of HyDE. In this section, we study the effect of changing their realizations. In particular, we consider smaller language models (LM) and fine-tuned encoders. We conduct our studies on TREC DL19/20.

5.1 Effect of Different Generative Models

In Table 4, we show HyDE using other instruction-following language models. In particular, we consider a 52-billion Cohere model (command-xlarge-20221108) and a 11-billion FLAN model (FLAN-T5-xxl; Wei et al. (2022)).[2] Generally, we observe that all models bring improvement to the unsupervised Contriever, with larger models bringing larger improvements. At the time when this paper is written, the Cohere model is still experimental without much detail disclosed. We can only tentatively hypothesize that training techniques may have also played some role in the performance difference.

5.2 HyDE with Fine-tuned Encoder

To begin with, HyDE with fine-tuned encoder is *not* the intended usage: HyDE is more powerful and irreplaceable when few relevance labels are present. Here we are interested to find out if and how HyDE embedding can affect fine-tuned encoders. In Table 4, we see that less powerful instruction LMs can negatively impact the overall performance of the fine-tuned retriever. (To remind our readers, Contriever^{FT} is in-domain supervisedly fine-tuned for TREC DL19/20). The performance degradations remain small. On the other hand, we also observe the InstructGPT model able to further bring up the performance, especially on DL19. This suggests that there may still exist certain factors not captured by the fine-tuned encoder but only by the generative model.

6 Conclusion

At the end of the paper, we encourage the readers to take a moment and reflect on the HyDE model. Compare it to some of the other recently seen retrievers or re-ranker. These other models probably differ in their architecture, training method, and/or task, but probably all of them involve modeling relevance scores between a pair of query and docu-

[2]Model sizes are from https://crfm.stanford.edu/helm/v1.0/?models.

ment. Dense retrievers consider vector similarities while self-attentive re-rankers regression scores. In comparison, the concept of relevance in HyDE is captured by an NLG model and the language generation process. We demonstrate in many cases, HyDE can be as effective as dense retrievers that learn to model numerical relevance scores. So, is numerical relevance just a statistical artifact of language understanding? Will a weak retriever theoretically suffice as the NLU & NLG models rapidly become stronger? Rushing to conclusions is not smart; more works need to be done to get answers. With this paper, we just want to raise these questions.

Concretely in this paper, we introduce a new paradigm of interactions between LLM and dense encoder/retriever. We demonstrate (part of) relevance modeling and instruction understanding can be delegated to the more powerful and flexible LLM. As a consequence, the need for relevance labels is removed. We are excited to see how this can be generalized further to more sophisticated tasks like multi-hop retrieval/QA and conversational search.

We argue HyDE is also of practical use though not necessarily over the entire lifespan of a search system. At the very beginning of the life of the search system, serving queries using HyDE offers performance comparable to a fine-tuned model, which no other relevance-free model can offer. As the search log grows, a supervised dense retriever can be gradually rolled out. As the dense retriever grows stronger, more queries will be routed to it, with only less common and emerging ones going to HyDE backend.

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A Appendix

A.1 Instructions

A.1.1 Web Search

Please write a passage to answer the question
Question: [QUESTION]
Passage:

A.1.2 SciFact

Please write a scientific paper passage to support/refute the claim
Claim: [Claim]
Passage:

A.1.3 Arguana

Please write a counter argument for the passage
Passage: [PASSAGE]
Counter Argument:

A.1.4 TREC-COVID

Please write a scientific paper passage to answer the question
Question: [QUESTION]
Passage:

A.1.5 FiQA

Please write a financial article passage to answer the question
Question: [QUESTION]
Passage:

A.1.6 DBPedia-Entity

Please write a passage to answer the question.
Question: [QUESTION]
Passage:

A.1.7 TREC-NEWS

Please write a news passage about the topic.
Topic: [TOPIC]
Passage:

A.1.8 Mr.TyDi

Please write a passage in Swahili/Korean/Japanese/Bengali to answer the question in detail.
Question: [QUESTION]
Passage: